Operating System

LAB # 5

Objective

"Implementation of Process and thread (Life cycle of process): (i) Process creation and Termination; (ii) Thread creation and Termination"

Processes and threads are fundamental concepts in operating systems.

What is a Process?

A process is an instance of a program that is being executed. It contains the program code and its current activity. Each process has its own memory space and resources.

Life Cycle of a Process

A process goes through various states during its lifetime:

1. New: The process is being created.
2. Ready: The process is waiting to be assigned to a processor.
3. Running: Instructions are being executed.
4. Waiting: The process is waiting for some event to occur (like I/O completion).
5. Terminated: The process has finished execution.

Creation of a Process

1. System Initialization
2. User Request

Termination of a Process

1. Normal Completion
2. Errors
3. Forced Termination

What is a Thread?

A thread is the smallest unit of execution within a process. A process can have multiple threads, all of which share the same memory space but execute independently.

Life Cycle of a Thread

Threads have a similar life cycle to processes but are generally simpler:

1. New: The thread is created.
2. Ready: The thread is ready to run and waiting for CPU time.
3. Running: The thread is executing.
4. Blocked: The thread is waiting for a resource or event.
5. Terminated: The thread has finished execution.

Creation of a Thread

1. Thread Libraries: Using thread libraries like POSIX Threads (pthreads) in Unix or std::thread **in C++11 and above.**

Termination of a Thread

1. Normal Exit
2. Canceled by Another Thread: Another thread can cancel it using functions like pthread_cancel() **in Unix.**
3. **Thread Exit Function: Explicitly calling thread exit functions like** pthread_exit() **in Unix.**

How to check Number of Threads:

lscpu

$$\text{Total Threads} = (\text{Thread(s) per core}) \times (\text{Core(s) per socket}) \times (\text{Socket(s)})$$

cat /proc/cpuinfo | grep processor | wc -l

Basic Commands

Locating executables files

which

Finding binary and source files for commands

whereis

Create file using touch command

touch

Important Flag
  -t (For Data and Time)

touch -t YYMMDDHHMM filename1

Create a thread and display numbers from 1 to 10;

<u>Before the code</u>

gedit fileName.c

```
#include<stdio.h>

#include<pthread.h>

#include <unistd.h>

void* thread_function(void* args)

{int i=0;

for(i=0;i<10;i++){

printf("%d\n",i);

sleep(1);

}

}

int main()

{

pthread_t id;

printf("Before thread is created \n");

int ret=pthread_create(&id,0,&thread_function,0);
```

pthread_join(id,0);

printf("After thread is executed\n");

return 0;

}

Note: First Close the file then try below commands

<u>After the code</u>

chmod +x fileName.c

gcc -o fileName fileName.c -lpthread

./fileName

- **gcc is the compiler command.**
- **thread.c is the name of c program source file.**
- **-o is option to make object file (o/p file or executable file).**
- **thread is the name of object file.**
- **-lpthread is option to execute pthread.h library file.**